



Bansilal Ramnath Agarwal Charitable Trust's

Vishwakarma Institute of Technology, Pune-37

(An autonomous Institute of Savitribai Phule Pune University)



Department of Engineering Science and Humanities (DESH)

Lab Manual

Course Code	Course Name	Teaching Scheme (Hrs. /Week)	Credits
ET1012	Applied Electromechanics	Theory: 3 Hours / Week Lab: 2 Hours / Week	4

Course Outcomes:

The student will be able to -

1. interpret various terminologies with reference to electromechanical systems such as robots.
2. distinguish and analyze working mechanisms of different actuators in electromechanical systems.
3. identify different end effectors and control.
4. understand application of electronic devices in digital and analog electronics
5. Illustrate the internal configuration, architecture, and terminologies of microcontroller.
6. interface different sensors with microcontroller-based systems

Class: - FY

Branch: - All Branches

Year: - 2024-25

Prepared By: D E S H

Required S/W and H/W: Open-source Arduino Software (IDE), Arduino Uno board, CPU with Pentium 5 or above.

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Experiment No. 1

Title of the assignment – _____

Name of the student – _____

Roll No. – _____ **Batch –** _____ **Division –** _____ **PRNo. –** _____

Date of performance – _____

Experiment No. 1

TITLE: Introduction to mechanical components

AIM: Study of the basic Mechanical components in robotics

APPARATUS: All Mechanical components.

- A) **Nut – Bolt and Screw:** As shown in figure 1.1 a **nut** is a type of [fastener](#) with a [threaded](#) hole. Nuts are almost always used in conjunction with a mating [bolt](#) to fasten multiple parts together. A **bolt** is a form of [threaded fastener](#) with an external male thread. Bolts are closely related to [screws](#). The two partners are kept together by a combination of their threads' [friction](#) (with slight [elastic deformation](#)), a slight [stretching](#) of the bolt, and [compression](#) of the parts to be held together.

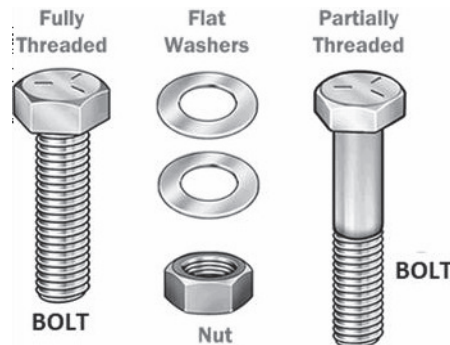
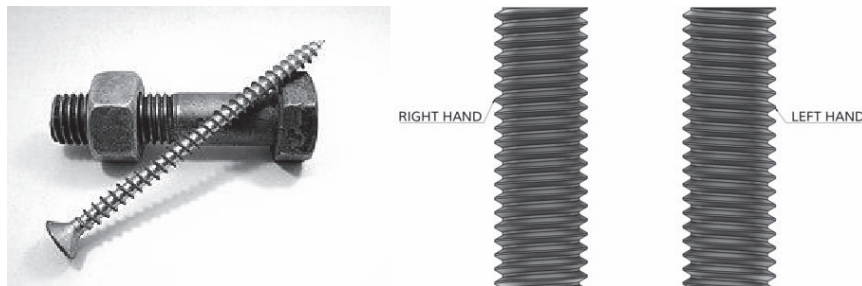


Figure 1.1 “Nut & Bolt”

As shown in figure 1.2 - a **screw** is a type of [fastener](#), in some ways similar to a [bolt](#), typically made of metal, and characterized by a [helical](#) ridge, known as a *male thread* (external thread). Screws are used to fasten materials by digging in and wedging into a material when turned, while the thread cuts grooves in the fastened material that may help pull fastened materials together and prevent pull-out. There are many screws for a variety of materials; those commonly fastened by screws include wood, sheet metal, and plastic.



(a) Screw compared to Nut-Bolt

(b) Right hand and Left-hand Threads

Figure 1.2 “Screw and type of threads”

- B) **Shaft:** In machinery, the general term “shaft” refers to a member, usually of circular cross-section as shown in figure 1.3, which supports gears, sprockets, wheels, rotors, etc., and which is used to transmit power from one part to another or from a machine which produces power to a machine which absorbs power. The material used for is alloy steel such as nickel chromium or chromium vanadium.



Figure 1.3 "Example of Shaft and key"

C) **Keys:** A key is a machine element as shown in figure 1.3 and 1.4 is fitted in an axial direction into the mating member such as pulleys, gears. The primary function of key is to prevent the relative rotation between the shaft and the mating member. Sunk keys are having keyways both in the shaft and the hub. These keys are suitable for heavy duty since they rely on positive drive. It is uniform in width and thickness throughout. There are two types of sunk keys as Square and Rectangular keys.

i) **Square key:** A square key is sunk half in the shaft and half in the hub of a gear, pulley or crank. In general, a square key has its sides equal to one-fourth of the shaft diameter.

ii) **Rectangular key:** Rectangular key is also called as the flat key. It is used where the weakening of the shaft is serious and where added stability of the connection is desired. Flat keys with uniform cross section are called Feather keys. Feather key permits axial movement of the hub on the shaft. There is one more type of the flat keys available called as Gib head flat key. Where it is provided with a Gib head which facilitates their removal.

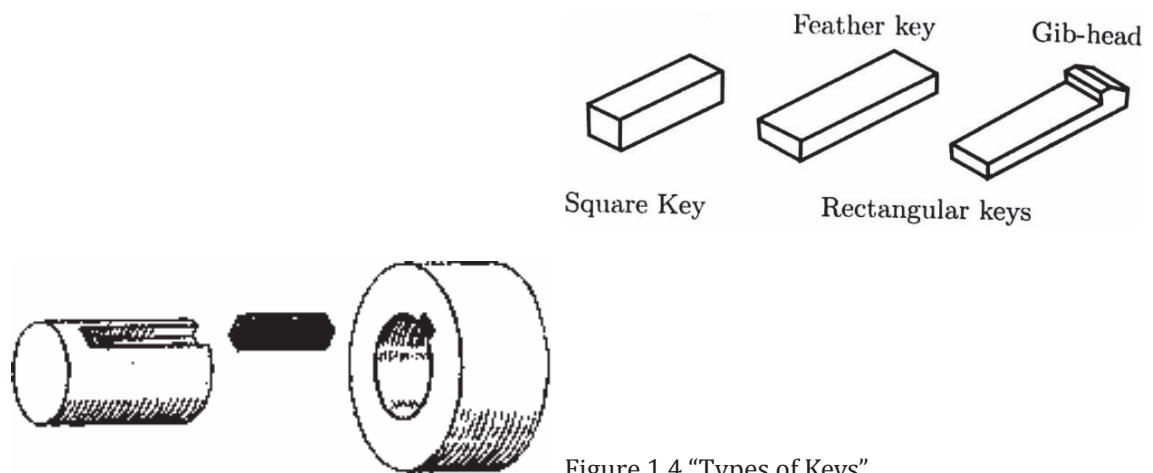


Figure 1.4 "Types of Keys"

D) **Couplers:** It is a mechanical device that joins two rotating shafts to each other for the purpose of transmitting power and absorb the misalignment between them. Couplers are generally categorized in the following varieties.

i) **Rigid Couplings:** Rigid Couplings are mainly used in areas where the two shafts are coaxial to each other. There are following types of couplings that fall under the rigid coupling's category.

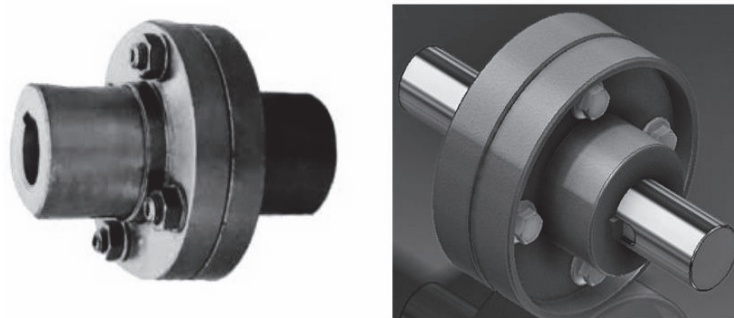
a) **Rigid Sleeve or Muff Couplings** – This is the basic type of coupling. This consists of a pipe whose bore is finished to the required tolerance based on the shaft size. Based on the usage of the

coupling a keyway is made in the bore in order to transmit the torque by means of the key. Two threaded holes are provided in order to lock the coupling in position. The figure 1.5 shows a type of the rigid sleeve or muff coupling.



Figure 1.5 “Muff Coupling”

b) *Flanged Coupling* – The coupling basically consists of two flanged end pieces as shown in the figure 1.6. The flanges are connected firmly by means of fitted bolts which are tightened accordingly to the torque to be transmitted. Further there are 02 types of flanges couplings as protected and un protected.



(a) Unprotected flange coupling (b) Protected Flange Coupling

Figure 1.6 “Types of Flange Couplings”

E) **Gears:** A gear is a kind of machine element in which teeth are cut around cylindrical or cone shaped surfaces with equal spacing. By meshing a pair of these elements, they are used to transmit rotations and forces from the driving shaft to the driven shaft. Gears are machine elements that transmit power by means of successively engaging teeth. Smaller Gear of the Pair is known as Pinion and Larger is known as Gear.

According to the position of axes of the shafts, there are following types of gears:

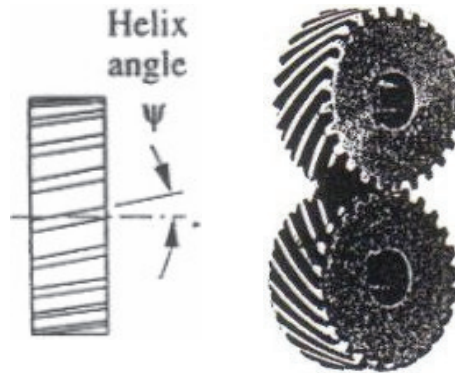
- a. Parallel
 1. Spur Gear
 2. Helical Gear
 3. Rack and Pinion
- b. Intersecting
 1. Bevel Gear
- c. Non-intersecting and Non-parallel
 1. Worm and worm gears

i) **Spur Gear:** in spur gears as shown in figure 1.7, the teeth are parallel to the axis of the gear. These are cost effective, easy to manufacture. But, spur gears only work with mating gear and it needs to have the axis of each gear parallel.



Figure 1.7 “Spur Gear”

ii) **Helical Gear:** In Helical gears as shown in figure 1.8 teeth are at an angle to the gear axis (usually 10° to 45°) hence called helix angle (see figure 1.6-a). Helical gears are smooth and quiet due to gradual tooth engagements. More tooth engagement allows greater power transmission for given gear size. But, Helical gears are more expensive and results in axial thrust.



a) Helix angle b) Parallel axis helical gear

Figure 1.8 “Helix angle and type of helical gears”

iii) **Bevel Gear:** Bevel gears are very similar to spur gears except they are intended to transfer rotation through a 90 degree translation. The bevel gears make a 90 degree transition easy but it does require a strong gear box. The speed and torque are treated the same in a bevel gear as they are in a spur gear.



Figure 1.9 “Bevel Gear”

iv) **Rack & Pinion:** Rack and pinion, mechanical device consisting of a bar of rectangular cross section (the rack), having teeth on one side that mesh with teeth on a small gear (the pinion). The pinion may have straight teeth, as in the figure, or helical (twisted) teeth that mesh with teeth on the rack that are inclined to the pinion-shaft axis.

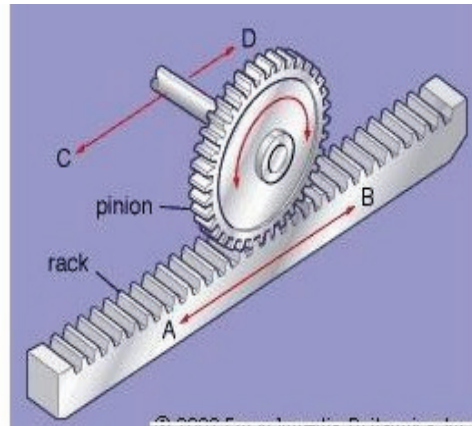


Figure 1.10 “Rack and pinion mechanism”

If the pinion rotates about a fixed axis, the rack will translate; i.e., move on a straight path, as shown by the arrow AB in the Figure. Some automobiles have rack-and-pinion drives on their steering mechanisms that operate in this way.

If the rack is fixed and the pinion is carried in bearings on a table guided on tracks parallel to the rack, rotation of the pinion shaft will move the table parallel to the rack as shown by the arrow CD in the Figure. On machine tools, rack-and-pinion mechanisms are used in this way to obtain rapid movements of worktables; the pinion shaft is usually rotated with a hand crank.

v) **Worm and Worm Gears:** A worm drive as shown in figure 1.11 is a gear arrangement in which a worm (which is a gear in the form of a screw) meshes with a worm gear (which is similar in appearance to a spur gear). The two elements are also called the worm screw and worm wheel. Worm gear has self-locking property.



Figure 1.11 “Example of Worm and Worm Gear”

- F) **Chain & Sprocket:** Sprockets and chain as shown in figure 1.12 can be used to change the speed, torque, or original direction of a motor. For sprockets and chain to be compatible with each other they must both have the same thickness and pitch. For the sprockets and chain to work effectively, all the sprockets should be on parallel shafts with their corresponding teeth on the same plane.

A **sprocket** is a toothed wheel that fits onto a shaft. It is prevented from rotating on the shaft by a key that fits into keyways in the sprocket and shaft.

A **chain** is used to connect two sprockets. One sprocket is the driver sprocket. The other sprocket is the driven sprocket. Motion and force can be transmitted via the chain from one sprocket to another, therefore from one shaft to another. Chains that are used to transmit motion and force from one sprocket to another are called **power transmission chains**.



Figure 1.12 “Chain and Sprocket Mechanism”

- G) **Belt Drives:** A belt is a looped strip of flexible material used to mechanically link two or more rotating shafts. A belt drive offers smooth transmission of power between shafts at a considerable distance. **Belt drives** are used as the source of motion to transfer to efficiently transmit power or to track relative movement. In a two-pulley system, depending upon the direction the belt drives the pulley, the belt drives are divided into two types. They are open belt drive and crossed belt drive.

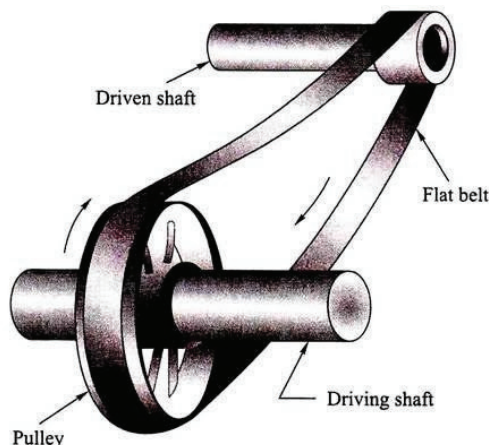


Figure 1.13 “Open and Cross Belt Drives”

As shown in figure 1.13 An open belt drive is used to rotate the driven pulley in the same direction of driving pulley. A crossed belt drive is used to rotate driven pulley in the opposite direction of driving pulley. Belt drives are simple and very economical. They don't need parallel shafts. They are lubrication-free. They require less maintenance cost. Belt drives are highly efficient in use (up to 98%, usually 95%). They are very economical when the distance

between shafts is very large. On the other hand, in belt drives Heat build-up occurs. Speed is limited to usually 35 meters per second. Power transmission is limited to 370 kilowatts. Operating temperatures are usually restricted to -35 to 85°C . Figure 1.14 shows some types of highly efficient belt drives.



Figure 1.14 “High - Efficiency Belt Drives”

H) **Bearings:** Mechanical systems have moving parts that transmit motion and force. The system may also convert one type of motion into another type of motion, e.g. rotary motion into reciprocating motion. In the ideal mechanical system, all the input motion and force is converted into output motion and force. However, friction between various parts of the mechanism results in heat being generated, wearing of surfaces in contact and loss of efficiency of the mechanical system. Lubricants and bearings are used to counteract the effects of friction in mechanisms.

Bearings are used to reduce the effects of friction. There are three main categories,

1. Bush bearing
2. Ball Bearing
3. Roller Bearing



Figure 1.15 “Types of Bush Bearings”

As shown in figure 1.15, **plain bearing**, also called bushes, consist of a tube-like sleeve. Some plain bearings have a flange at one end. Plain bearings are made from materials that have a low coefficient of friction.



Figure 1.16 “Ball Bearings”

As shown in figure 1.16 **ball bearing** is a type of rolling-element bearing that uses balls to maintain the separation between the bearing races. The purpose of a ball bearing is to reduce rotational friction and support radial and axial loads. It achieves this by using at least three races to contain the balls and transmit the loads through the balls. In most applications, one race is stationary and the other is attached to the rotating assembly (e.g., a hub or shaft). As one of the bearing races rotates it causes the balls to rotate as well. Because the balls are rolling they have a much lower coefficient of friction than if two flat surfaces were sliding against each other.



Figure 1.17 “Roller Bearings”

As shown in figure 1.17 roller bearing consist of a number of truncated conical rolling elements held in a cage between the inner and outer races. Taper roller bearings have a high load bearing capacity due to the line contact between each of the roller bearings and the inner and outer races. Taper roller bearings can support radial and axial loads. This conical geometry creates a linear contact patch which permits greater loads to be carried than with spherical (ball) bearings, which have point contact. The geometry means that the tangential speeds of the surfaces of each of the rollers are the same as their raceways along the whole length of the contact patch and no differential scrubbing occurs

1) **Springs:** A spring is a mechanical device which is used for efficient storage and release of energy. It is defined as an elastic body whose function is to distort when loaded and to recover its original shape when the load is removed. Helical spring is a spiral wound wire with a constant coil diameter and uniform pitch. Function of helical spring are,

- To apply force in valves brakes and clutches
- To absorb shock Used to store energy and subsequently release it.
- e.g. clock springs, toys
- For supporting moving masses
- To maintain a force between contacting surfaces e.g. Cam

a) **Compression spring:** These are designed to operate with a compressive load and found in shock absorbers, spring mattresses, mechanical pencils, and retractable pens, ball pens as shown in figure 1.18.

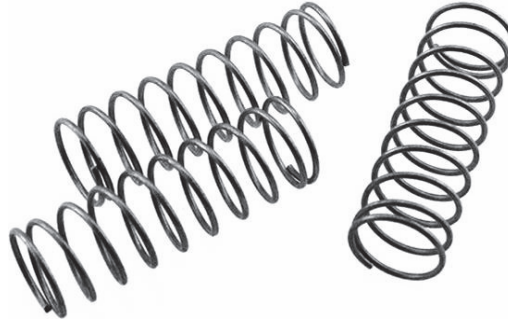


Figure 1.18 “Compression Spring”

b) **Extension spring:** These are designed to operate with a tensile load. An archetypical example is a Slinky, but these are also found in luggage scales and garage door mechanisms. A typical example is bicycle stand spring as shown in figure 1.19.



Check List of components to observe

SN	Component	Location (where it is observed)	SN	Component	Location (where it is observed)
1	Nut Bolt		15	Rake and Pinion	
2	Screws		16	Bevel Gear	
3	Rivets		17	Warm Gear	
4	Cotter Joint		18	Flat Belt	
5	Pin Joint		19	V-Belt	
6	Shaft		20	Rope	
7	Spindle		21	Pulleys	
8	Keys		22	Washers	
9	Couplings		23	Helical Spring	
10	Washers		24	Leaf Spring	
11	Spanners		25	Eye Bolt	
12	Bearings		26	Dome Nut	
13	Spur Gear		27	Wing Nut	
14	Helical Gear		28	Cam Follower	

Conclusion:

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Vishwakarma Institute of Technology
Department of Engineering, Sciences and Humanities
FY: 2024-2025: Semester I & II
ET1012: Applied Electromechanics



Experiment No. 2

Title of the assignment – _____

Name of the student – _____

Roll No. – _____ Batch – _____ Division – _____ PRNo. – _____

Date of performance – _____

Experiment 2

Introduction to electrical and electronic components

AIM: To Study basic electrical and electronic components used in an electromechanical system such as robotics.

APPARATUS: Various types of wires, cables, switches, sockets, fuses, lugs, resistors, capacitors, diodes, Light Emitting Diodes, Infra-Red LEDs, transistors, relay, connectors, cables, Digital Multi Meter, breadboard.

Theory:

- 1. Wires :** There are many types of wires among which P.V.C. (Poly Vinyl Chloride) wires are most used. These have conductors with P. V. C. insulation. P. V. C. is non-hygroscopic, tough, durable, resistant to corrosion and chemically inert. However, P. V. C. softens at high temperature and hence is not used where extremes of temperature occur. e.g. in heating appliances.
- 2. Cables :** These consist of several individually insulated conductors, which are put together inside a protective mechanical covering. There can be two types of cables: 1) Aerial cables 2) Underground cables. The construction of a typical cable is shown in figure below.

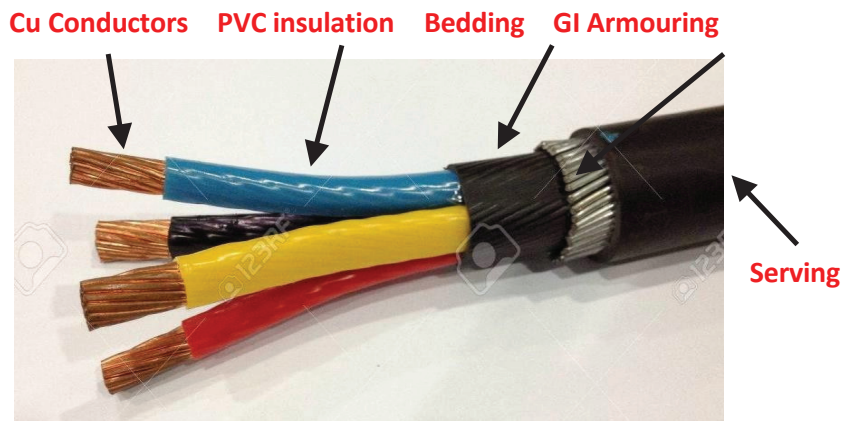


Figure 2.1 “Construction of a cable”

- 3. Switches :** These are used to make the circuits ON or OFF. The different types are -
 - 1) SPST (single pole single throw)
 - 2) SPDT (single pole double throw) also called Two-way switch

- 3) DPDT (double pole double throw)
- 4) DPST (double pole single throw)
- 5) SPDT with center OFF
- 6) Intermediate switch
- 7) Emergency switch
- 8) Push button switch
- 9) Roller type switch
- 10) Hinge lever type switch
- 11) Plunger type switch

For various types of switches refer figure 2.2.



Piano Switch



Emergency Switch



Plunger type Switch

Figure 2.2 “Types of switches”

- 4. Sockets :** These are the points from which electricity can be tapped. Generally, sockets have three terminals namely i) phase ii) neutral iii) earth. Sockets can be classified depending upon the current carrying capacity such as 5 Amp or 15 Amp. or upon single-phase, three- phase (industrial sockets) etc. Refer figure 2.3.



Figure 2.3 “Socket”

- 5. Fuses :** A fuse is a device, which disconnects and protects the circuit from short circuit and overload. It consists of a porcelain base with a fuse wire, which melts at high

temperature. When excess current flows, the fuse wire melts and disconnects the circuit from the supply. Two types of the fuses are available.

1. Rewirable fuse
2. H. R. C. fuse (High Rupturing capacity). (Refer figure 2.4)



Figure 2.4“HRC fuses”

The rewirable fuse is also called as Kit Kat fuse. The types of H. R. C. fuse are Pin type, Bottle type and Screw type. Rewirable fuses are cheap and HRC fuses are costly. But they are very much useful in hazardous area like Chemical plants, LPG filling plants.

6. Lugs

Lugs are used to make firm connection of wires or cables and to increase the life of joints. The lugs can be classified as

3. Eye or hole type
4. U type,
5. Pin type.

Generally, copper or aluminum lugs are used. Many times, they are silver coated, as silver is the best conductor. A crimping tool is used to make a firm connection of a wire and lug. Nut bolting is used to make a firm joint of two lugs. For various types of lugs, refer figure 2.5



Figure 2.5“Different types of lugs”

- 7. Resistors:** Resistors offer a resistance to the flow of current. There are two types of resistors – fixed and variable. Potentiometer is the variable resistor which is used to vary the resistance by rotating the shaft. For various types of resistors, refer figure 2.6.



Figure 2.6“Different types of resistors”

How to calculate the value of a resistor? Resistor value is calculated by the color code. Table 2.1 illustrates the color codes of resistor.

Color	1 st Band	2 nd Band	3 rd Band	Decimal Multiplier	Tolerance
Black	0	0	0	1	
Brown	1	1	1	10	±1%
Red	2	2	2	100	±2%
Orange	3	3	3	1K	
Yellow	4	4	4	10K	
Green	5	5	5	100K	
Blue	6	6	6	1M	
Violet	7	7	7	10M	
Gray	8	8	8	100M	
White	9	9	9	1000M	
Gold				0.1	±5%
Silver				0.01	±10%
None					±20%

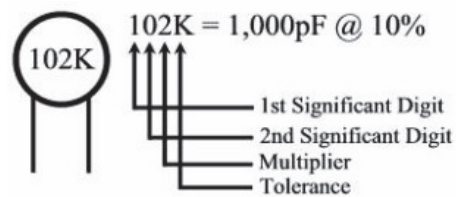
Table 2.1“Color codes of resistors”

8. Capacitors: Capacitors are used to store the charge. Capacitors can be fixed or variable. In robotics, mostly fixed capacitors are used. The capacitors are available with or without polarity. Ceramic and mica capacitors are of no-polarity and electrolytic capacitors come with polarity. Various types of capacitors are shown in figure 2.7.



Figure 2.7“Different types of capacitors”

How to calculate the value of ceramic capacitor? For example, what is the value for 102k ceramic capacitors?



Code	Tolerance
C	±0.25pF
J	±5%
K	±10%
M	±20%
D	±0.5pF
Z	80% / -20%

Figure 2.8“Identification of capacitor value from digit codes”

Capacitances vary from 22pF to about 15000μF. Values <0.1μF are mainly mica and ceramic capacitors and values C>= 1μF are electrolytic capacitors.

9. Diodes: Diode is the basic two terminal component that allows the flow of current only in one direction. Most diodes are made of semiconductor materials like silicon, germanium, or selenium. Diodes can be used as rectifiers, voltage regulators, signal limiters, switches, signal modulators and oscillators. In robotics, mainly, LEDs, IR LEDs and photo diodes are used. Regular diodes and Zener diodes are rarely used. Various types of diodes are shown in figure 2.9.



Figure 2.9“P-N junction, Zener, and Light Emitting Diodes”

10. Transistors: A transistor is a semiconductor device that exhibits all the properties of a switch, allowing or blocking the flow of electrons. It has three terminals, one for input, one for output and one for controlling switching. It is the fundamental building block of modern electronic devices and is commonly found in circuit boards as discrete parts or embedded into integrated circuits. Refer figure 2.10 for symbol and various types of transistors.



Figure 2.10“Symbol and types of transistors”

11. Relays: A relay is a device that opens or closes the contacts to cause the operation of the other electric control. It detects the intolerable or undesirable condition with an

assigned area and gives the commands to the circuit breaker to disconnect the affected area. Thus

protects the system from damage. The pole and throws are the configurations of the relay, where the pole is the switch, and the throw is the number of connections. The single pole, the single throw is the simplest type of relay which has only one switch and only one possible connection. Similarly, the single pole double throw relay has a one switch and two possible connections. Figure 2.11 shows standard relay.

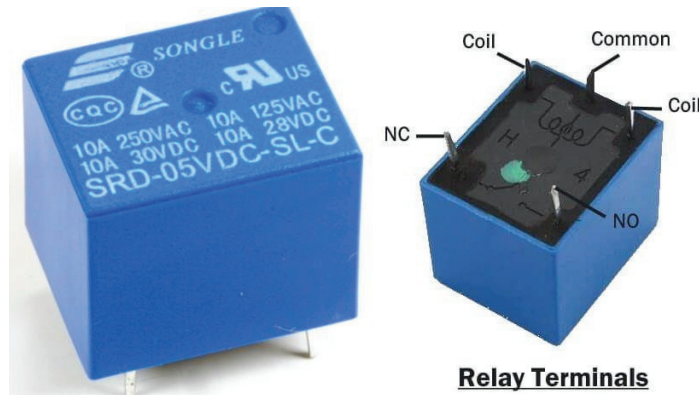


Figure 2.11“Relay and relay terminals”

12. Connectors: An electronic connector is an electro-mechanical device whose purpose is to quickly and easily disconnect or interrupt a circuit path. Connectors come in a variety of sizes, shapes, complexities, and quality levels. Their function dictates their design, and different features are added to adjust the ease of connection, mating type, durability, insulation between pins, etc. In addition, because many connectors must perform their job in rugged conditions, their construction is often adjusted to provide protection from vibrations, extreme temperatures, dirt, water, contaminants, and more. Refer figure 2.12 for various connectors.



Figure 2.12“Types of connectors”

13. Cables: Cable is a conductor or group of conductors for transmitting electric power or telecommunication signals from one place to another. Electric communication cables transmit voice messages, computer data, and visual images via electrical signals to telephones, wired radios, computers etc.

14. Digital Multi Meter: A digital multi meter or DMM is a test tool used to measure two or more electrical values—principally voltage (volts), current (amps) and resistance (ohms). It is a standard diagnostic tool for technicians in the electrical/electronic industries.

The face of a digital multi meter typically includes four components:

1. Display: Where measurement readouts can be viewed.
2. Buttons: For selecting various functions; the options vary by model.
3. Dial (or rotary switch): For selecting primary measurement values (volts, amps, ohms).
4. Input jacks: Where test leads are inserted.



Figure 2.13 Digital Multi Meter

15. Breadboard: A breadboard is a solder less device for temporary prototype with electronics and test circuit designs. Most electronic components in electronic circuits

can be interconnected by inserting their leads or terminals into the holes and then making connections through wires where appropriate. The breadboard has strips of metal underneath the board and connects the holes on the top of the board. Figure 2.14 shows breadboard.



Figure 2.14“Breadboard”

Check List of components to observe

SN	Component	Location (where it is observed)	SN	Component	Location (where it is observed)
1	Switches		10	LEDs	
2	Sockets		11	Resistors	
3	3 Phase Cables		12	Potentiometer	
4	Fuse		13	Capacitor	
5	MCBs		14	Push Button	
6	Logs		15	Multimeter	
7	3 Phase Socket		16	Zener Diode	
8	Motor		17	Relays	
9	Rheostat		18	Breadboard	

Conclusion:



Experiment No. 3

Title of the assignment – _____

Name of the student – _____

Roll No. – _____ **Batch –** _____ **Division –** _____ **PRNo. –** _____

Date of performance – _____

Experiment No. 3

TITLE: Visit and demonstration of a robot

AIM: To study Pick and Place operation using SCARA robot

APPARATUS: Parts of SCARA Robot

1) Robot Structure

Line Diagram of Complete system

1. Manipulator (Figure and Explanation)

2. Controller (Figure and Specifications)

3. End Effector (Figure and Explanation)

Functions of the robot.

- 1) Pick and place the color blocks from Left portion to right
- 2) Programming the robot for the said sequence.

Procedure.

1. Switching ON the robot

- i. Plug in the main Power supply cable (230 V 6A).
- ii. Check the MCB and RCCB are in ON position.
- iii. Switch ON the Main power supply and wait till the stabilizer gets ON. (If the stabilizer is turned off and turned ON in very quick succession then it may take 60 sec to 90 sec for stabilizer to start).
- iv. Turn ON the rotary ON/OFF switch so the robot gets ON.
- v. Emergency switch on Control Panel and Teach Pendant should be in released condition.

2. Operating Robot in Manual mode (teach pendant)

- i. Turn the Auto/Manual selector switch on control panel to manual mode.
- ii. Turn ON the T/B enable key (Back side of the teach pendant)
- iii. Pull the safety lever on the backside of the T/B slightly so that the lever is neither completely pressed nor completely released and you hear a single click noise. (The safety lever needs to be kept pulled every time for moving the robot.)
- iv. Press "SERVO ON" button to turn on the servos.
- v. Press jog mode button to move the robot. • Check the status of robot mode (XYZ or Joint) at the left top corner of the display.
 - This mode can be changed by pressing F1 key.
 - For moving the robot in Joint mode Press key X, Y, Z, A for joint 1,2,3,4 respectively.
 - For moving the robot in XYZ mode Press key X, Y, Z for X axis, Y axis, Z axis, respectively.



Check List of components to Demonstrate

SN	Component	Remark
1	Base	
2	Joints	
3	Linkages	
4	Controls	
5	Manual Handling	
6	Pick and Place	
7	Joystick working	
8	Gripper details	
9	X,Y,Z Travel	
10	DOF	

Conclusion:



Experiment No. 4

Title of the assignment – _____

Name of the student – _____

Roll No. – _____ Batch – _____ Division – _____ PRNo. – _____

Date of performance – _____

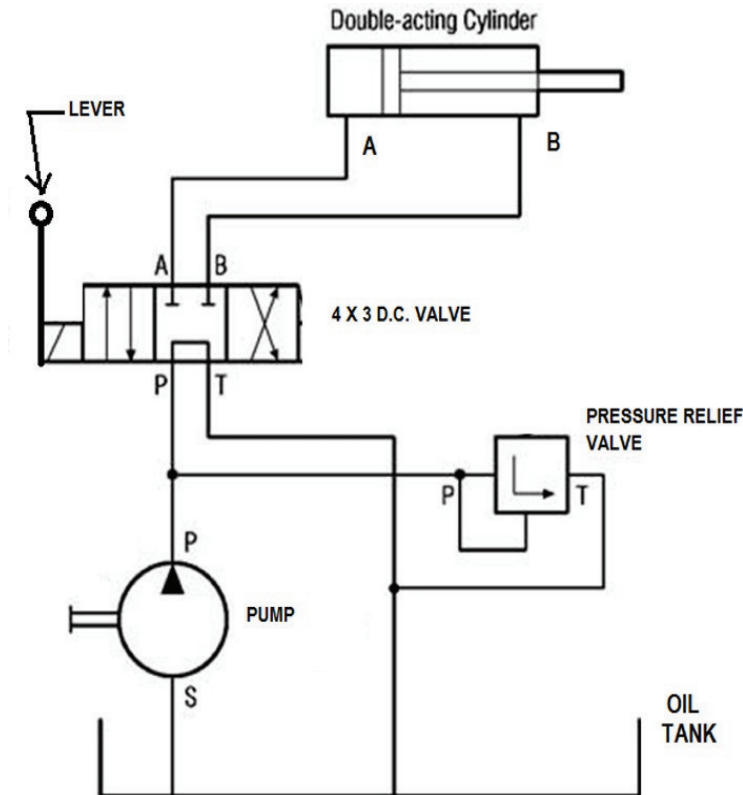
Experiment No. 4

TITLE: Demonstration of Hydraulic and Pneumatic Actuators

AIM: To Hydraulic and Pneumatic Circuit Components and working

APPARATUS: Oil tank, filter, pump, manually operated, 4x3 D.C. valve, double acting cylinder, hoses, hydraulic motor, Pneumatic Trainer etc.

Basic hydraulic circuit for double acting cylinder

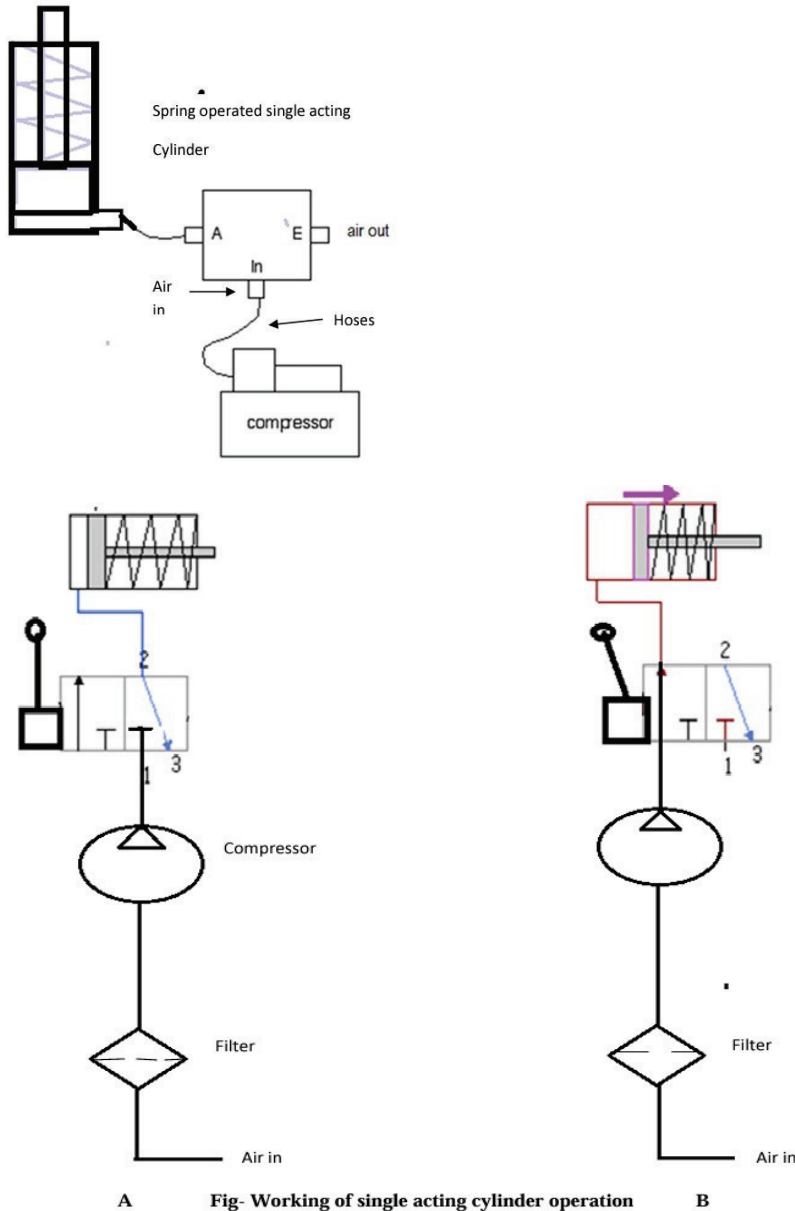


In this experiments It is to demonstrate that, how the pressure energy of oil converted in mechanical energy by using double acting cylinder, hydraulic motor. The circuit diagram to control double-acting cylinder is shown in Fig 1. The control of a double acting hydraulic cylinder is described as follows:

1. When the 4/3 valve is in its neutral position (tandem design), the cylinder is hydraulically locked and the pump is unloaded back to the tank.
2. When the 4/3 valve is actuated into the flow path, the cylinder is extended against its load as oil flows from port P through port A. Oil in the rod end of the cylinder is free to flow back to the tank through the four-way valve from port B through port T.
3. When the 4/3 valve is actuated into the right-envelope configuration, the cylinder retracts as oil flows from port P through port B. Oil in the blank end is returned to the tank via the flow path from port A to port T.

4. At the ends of the stroke, there is no system demand for oil. Thus, the pump flow goes through the relief valve at its pressure level setting unless the four-way valve is deactivated.

Basic pneumatic circuit for the working of single and double acting cylinder



Theory -

Connect all components as per above shown fig. high pressurized air enters in 3X2 Valve manually forward positions of lever air enters in cylinder, when lever operated backward the air exhausted in surrounding through port E due to force of compressed spring

Forward motion-Due to pressurized air enters in cylinder.

Backward motion- spring pressure applied on area of piston surface A] In fig. there is return movement of piston due to spring force. So, air is delivered (exhausted) to atmosphere. Here

3x2 direction control valve is used. The air is exhausted to atmosphere through port 2 to 3. B] In fig. there is forward movement of piston due to high pressure air. So, air is entered to cylinder. And apply the force on piston surface Here 3x2 direction control valve is used. The air is entered in cylinder through port 1 to 2.

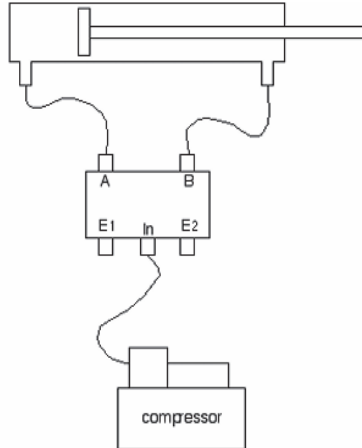


Fig. schematic diagram of double acting pneumatic cylinder

Tutorial 1.

Draw a Hydraulic Circuit to operate

- 1) Two double acting Cylinder Simultaneously**
- 2) Two Double acting cylinders one after the other.**

(Circuit and Explanation)



Tutorial 2.

Draw a Pneumatic Circuit for Robot pick and place mechanism

(Circuit and Explanation)

Observed components in Demonstration

SN	Component	Remark
1	Pump	
2	Reservoir	
3	Pressure Control Valve	
4	Direction Controls Valve	
5	Flow Controls Valve	
6	Hose Pipes	
7	Points A, B, P, and T	
8	Double acting Cylinder	
9	Pneumatic Circuit Parts	
10	FRL Unit	

Conclusion:

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Experiment No. 5

Title of the assignment – _____

Name of the student – _____

Roll No. – _____ Batch – _____ Division – _____ PRNo. – _____

Date of performance – _____

Experiment No-5.

Title: Know your Micro-controller kit.

Tasks:

- A) Get introduced to the Arduino Uno kit hardware.
- B) Get introduced to the features of Arduino IDE.

History & important features of the Arduino UNO kit hardware:

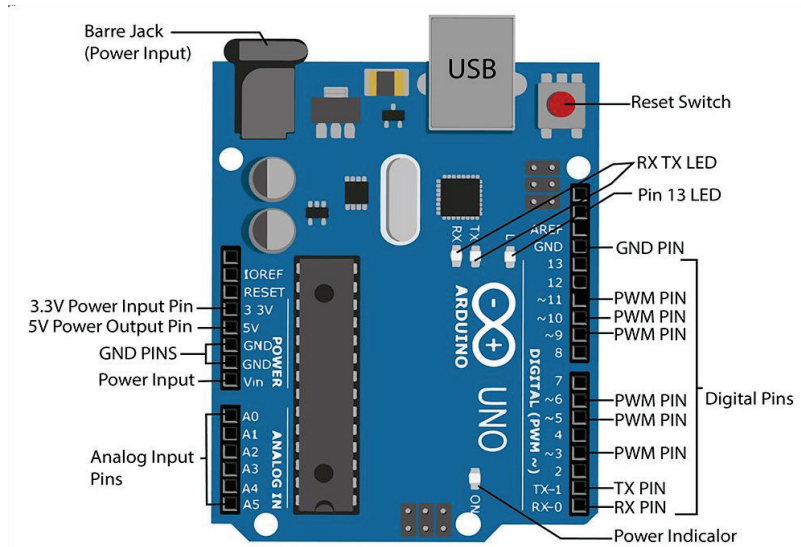
The Arduino UNO is an open-source microcontroller board based on the Microchip AT mega 328P microcontroller and developed by Arduino. Arduino is a prototype platform(open-source) based on an easy-to-use hardware and software. It consists of a circuit board, which can be programmed (referred to a microcontroller) and a ready- made software called Arduino IDE(Integrated Development Environment), which is used to write and upload the computer code to the physical board.

Key Features -

- Arduino boards are able to read analog or digital input signals from different sensors and turn it in to an output such as activating a motor, turning LED on/off, connect to the cloud and many actions.
- You can control your board functions by sending a set of instructions to the microcontroller on the board via Arduino IDE (referred to as uploading software).
- Unlike most previous programmable circuit boards, Arduino does not need an extra piece of hardware (called a programmer) in order to load a new code onto the board. You can simply use the USB cable since it provides the in system programming facility.
- Additionally, the Arduino IDE uses a simplified version of C++, making it easier to learn to program.
- Finally, Arduino provides a standard form factor that breaks the functions of the micro-controller into an easily accessible package.

Specifications of Arduino UNO:

- Microcontroller: ATmega328
- Operating Voltage – 5V Input voltage - 7-12 V Recommended), 6-20v (limits).
- Digital / pins: 14 C6 are PWM) 01 Analog Vo pins 16. • De current per 1/0 pin - 40mA for 3.3v pin-50mA
- Flash memory - 32 KD of which 0.5 XB wed by boot loader → SRAM -2kh (ATmega328)
- EEPROM - 1xb. (AT mega 32), Clock speed - 16 MHz



Parts of Arduino Uno:

Important features of Arduino IDE and the menu:

- Sketch editing tools- Arduino programs are called as sketch. Tool has various libraries to perform different mathematical, interfacing functions. Also provides the facility of a Serial Monitor to observe interim results. One can choose board of the choice as per requirements. Port selection is also available.
- Arduino programs can be divided in three main parts: Structure, Values (variables and constants), and Functions. In this tutorial, we will learn about the Arduino software program, step by step, and how we can write the program without any syntax or compilation error. Software structure consists of two main functions –
- Setup() function - The setup() function is called when a sketch starts. Use it to initialize the variables, pin modes, start using libraries, etc. The setup function will only run once, after each power up or reset of the Arduino board.
- Loop() function - After creating a setup() function, which initializes and sets the initial values, the loop() function does precisely what its name suggests, and loops consecutively, allowing your program to change and respond. Use it to actively control the Arduino board.

Pins of Arduino Uno:

Vin: This is the input voltage pin of the Arduino board used to provide input supply from an external power source.

5V: This pin of the Arduino board is used as a regulated power supply voltage and it is used to give supply to the board as well as onboard components.

3.3V: This pin of the board is used to provide a supply of 3.3V which is generated from a voltage regulator on the board

GND: This pin of the board is used to ground the Arduino board.

Reset: This pin of the board is used to reset the microcontroller. It is used to Resets the microcontroller.

Analog Pins: The pins A0 to A5 are used as an analog input and it is in the range of 0-5V.

Digital Pins: The pins 0 to 13 are used as a digital input or output for the Arduino board.

Serial Pins: These pins are also known as a UART pin. It is used for communication between the Arduino board and a computer or other devices. The transmitter pin number 1 and receiver pin number 0 is used to transmit and receive the data resp.

External Interrupt Pins: This pin of the Arduino board is used to produce the External interrupt and it is done by pin numbers 2 and 3.

PWM Pins: This pins of the board is used to convert the digital signal into an analog by varying the width of the Pulse. The pin numbers 3, 5, 6,9,10 and 11 are used as a PWM pin.

SPI Pins: This is the Serial Peripheral Interface pin; it is used to maintain SPI communication with the help of the SPI library. SPI pins include:

1. SS: Pin number 10 is used as a Slave Select
2. MOSI: Pin number 11 is used as a Master Out Slave In
3. MISO: Pin number 12 is used as a Master In Slave Out
4. SCK: Pin number 13 is used as a Serial Clock

LED Pin: The board has an inbuilt LED using digital pin-13. The LED glows only when the digital pin becomes high.

AREF Pin: This is an analog reference pin of the Arduino board. It is used to provide a reference voltage from an external power supply.

Conclusion:



Experiment No. 6

Title of the assignment – _____

Name of the student – _____

Roll No. – _____ Batch – _____ Division – _____ PRNo. – _____

Date of performance – _____

Experiment No- 6

Title: Interfacing of LED with Arduino Uno kit.

Pre-requisite: Information about safe voltage levels and protection of LEDS.

Tasks: Interface external LEDs with the UNO kit in a variety of combination and write a program.

- A) Blink 1 LED with predetermine timing for ON and OFF
- B) Blink 2 or 4 LED's with predetermine timing for ON and OFF (together and alternately)
- C) Blink minimum 7-8 LED's to switch ON and OFF sequentially in both directions
- D) To simulate working of traffic signal for two three way road. Formulate your own theme and delay cycle Decide the hardware required and assemble the same on a breadboard for your theme

Theory: As per the problem statement, we need to connect given LEDS into the specified variety of combinations using Arduino Uno, jumpers, LED's, breadboards and resistance.

In the case of 5-LED problem statement, we need to arrange the LEDs in Specified manner along with appropriate resistance in order to prevent LED from bursting. Similarly, vehicle turn indicator and traffic signal are safety measures, which would work on similar principle.

Working: When we complete the total connection and coding part, the working can be understood easily.

- First of all, in all the 3 cases, LED's blink or glow so using "Digital write" they can glow.
- In the alternating lightening case 1, 3, 5 glow while LED, 2, 4 are off and vice-versa...

Whereas in loop run, we just increase the delay time. In case of parking light system, similar mechanism is used. used along with delay, where the system depicts left led glowing to show that vehicle is turning left and right glower when the vehicle is turning right and bath of them glow when the vehicle u parked, while above two cases are similar, traffic light has 6 Led's in total (for two way) and two glow at a time, where delay time is quite different as compared to above two cares.



Electronic components and materials required:

- Arduino Uno - operating voltage 5.v. AT mega 328P. No of digital i/o pins-14, Input voltage limits 6-20 V.
- LEDs should be available in different colors as per requirement. Specification forward biased. 1.2 to 36 V rated about 10 to 30 mA.
- Bread board - 5A rated. 2 distribution strips.
- Jumpers should be male to male.
- Resistors - As per requirement, but 182 recommended.
- Arduino IDE software - Latest supporting version supporting UNO.

ACTUAL CONNECTED CIRCUIT PHOTOS

Task 1

Circuit Connection photo before operation/ Working

Actual Photograph during Working of the circuit

EXPALNATION OF THE RESULT OF EACH TASK

Industrial application:

- This project can be used in traffic signals and vehicle indicator.
- In day to day life, in festival lightings, road safety boards, etc.

Conclusion:



Experiment No. 7

Title of the assignment – _____

Name of the student – _____

Roll No. – _____ Batch – _____ Division – _____ PRNo. – _____

Date of performance – _____

Experiment No.-7

Title: Interfacing of LDR with ARDUINO UNO kit.

Pre-requisites: Assignment number 2 must be performed and Principle of working of LDR.

Tasks: Interface LDR with the UNO kit and write a program.

A) : Interface LDR with the UNO kit such that when you cover the LDR surface, an LED will glow indicating night hours. When uncovered, the LED should switch OFF.

B) To implement automated street light ON OFF system depending upon ambient light intensity. Use at least 3-LEDs as street Light.

B) To count number of objects passing through. The system should use light source (LED) and detector (LDR). Initially count should be Zero and with every object passing it should be incremented by '1' Display the count on serial monitor of the serial monitor.

Theory:

Working Principle-The working principle of an LDR is photoconductivity, which is nothing but an optical phenomenon. When the material absorbs the light then the conductivity of the material enhances. When the light falls on the LDR, then the electrons in the material's valence band are eager to the conduction band. But, the photons in the incident light must have energy superior to the band gap of the material to make the electrons jump from one band to another band (valence to conduction).

Hence, when light has ample energy, more electrons are excited to the conduction band which grades in many change canters. When the effect of this process and the flow of the current starts flowing more. the resistance of the device decreases.

Sensor Specifications -

- Resistance: 400 ohm to 400 kilo ohm.
- Normal resistance variance: 1 kilo ohm to 10 kilo ohm.
- Sensitivity: about 3 msec.
- Voltage rating: - 3-12 V.

For the first problem statement, an automated street light can be implemented. Here, when the light is falling on the LDR the LED will turn off and when the LDR is not receiving no / low light intensity the

LEDS will glow, which depicts that during the day the street lights would be off and during night times they will turn on.

For the second problem statement, a counted is to be implemented where in the system would record the number of objects posted through. Here, an LDR and & light source led would be placed opposite to each other. When an object passes between both of them it would block the light source to the LDR due to which the LDR will turn off hence incrementing the count. So, every time an object passes the count will be incremented and from this, we can get the total number of objects passed. The final count would be printed on the serial monitor.



Electronic components and materials required:

- 1) Arduino board (UNO): operating voltage 5V (AT mega 328P)
- 2) LDR Rating - 250 V
- 3) Bread board- 5A rated.2 distribution strips.
- 4) Arduino IDE software.
- 5) LEDES –lids as per requirement (4) and in different colors.
- 6) Jumpers - should be male to male.
- 7) Resistors as per requirement, but 1 kilo ohm recommended.

Interfacing Circuit diagram: (Draw Connections between Arduino Uno and the Sensor as well as other components)

The first Tasks was connecting the LDR and Arduino Uno on the bread board which was to be done very carefully to avoid errors. Then, the coding part expected was quite easy and lastly it was working at.



ACTUAL CONNECTED CIRCUIT PHOTOS

Task 1

Circuit Connection photo before operation/ Working

Actual Photograph during Working of the circuit

Photograph of Serial Monitor depicting the result

EXPALNATION OF THE RESULT OF EACH TASK

Application:

Apart from being used in street lights, LDR is used in the:

- Infrared astronomy.
- It is used in light failure alarm circuits and used in light meter.
- LDR is also used in smoke detectors.

Conclusion:



Experiment No. 8

Title of the assignment – _____

Name of the student – _____

Roll No. – _____ Batch – _____ Division – _____ PRNo. – _____

Date of performance – _____

Experiment No.-8

Title - Interfacing of ultrasonic sensor with Arduino Uno kit.

Pre-requisite: Principle of working of ultrasonic sensor and Calculation of distance

Tasks: Interface ultrasonic sensor with the UNO kit and write a program.

- A) To find distance (in cm) between the sensor and fixed surface like obstacle or wall etc. Display the distance on Serial monitor. Verify the distance by actual measurement.
- B) Give alert using Buzzer and LED. If distance goes above or below a predetermined level
- C) Ultrasonic distance between two ranges finding and turn ON appropriate LED

Theory:

Working Principle - Ultrasonic sensors work by sending out a sound wave at a frequency above the range of human hearing. The transducer of the sensor acts as a microphone to receive and send the ultrasonic sound.

Sensor Specifications:

- Operating voltage +5v
- Theoretical measuring distance - 2cm to 450 cm
- Practical measuring distance -2 cm to 80 cm
- Accuracy-3mm, measuring angle covered $< 15^\circ$
- Operating current < 15 mA.
- Operating frequency – 40 Hz.
- Buzzer

Firstly, the sensor and Arduino Uno are connected. Now, as per the problem statement, we have to measure the distance between the sensor and a fixed surface. Once the power supply is given, the sensor will send and receive the sound waves and calculate the time taken for the waves to return. Distance will be calculated by using the formula $\text{distance} = (\text{speed of sound} \times \text{time}) / 2$, and this distance measured will be displayed on the Serial monitor.



Operating components and materials required:

- Arduino board (und) - operating voltage 5V (At mega 328P)
- Breadboard 1-5A rated, 2-distribution strips.
- Jumpers - set of make to male connectors & a set of male to female connectors.
- Arduino IDE Software - updated to the latest version.
- Ultrasonic Sensor - operating voltage-5x
- Buzzer

ACTUAL CONNECTED CIRCUIT PHOTOS

Task 1

Circuit Connection photo before operation/ Working

Actual Photograph during Working of the circuit

Photograph of Serial Monitor depicting the result

EXPALNATION OF THE RESULT OF EACH TASK



Industrial application: Apart from the distance measurement, ultrasonic sensors are used in robot sensing, presence detection, tank level detection and many applications on the production line, etc.

Conclusion:



Experiment No. 9

Title of the assignment – _____

Name of the student – _____

Roll No. – _____ Batch – _____ Division – _____ PRNo. – _____

Date of performance – _____

Experiment No. 9

Title: 16x2 LCD Interfacing with Arduino Uno Kit

Prerequisites: Working Principle and Character formation logic of LCD

Tasks:

- A) Write a program to display Hello World! On LCD at desired location
- B) Write a program to scroll custom message on Right or Left direction on LCD.
- C) Write a program to display custom characters on LCD
- D) Display distance measured by ultrasonic sensor.

Theory:

Liquid Crystal Display (LCD) is the display device that can be interfaced with Arduino. For this, The Liquid Crystal library (Liquid Crystal.h) allows to control LCD displays. LCDs have a parallel interface, meaning that the microcontroller has to manipulate several interface pins at once to control the display. The interface consists of the following pins:

- 1) A **register selects (RS) pin** that controls where in the LCD's memory for writing data to. One can select either the data register, which holds what goes on the screen, or an instruction register, which is where the LCD's controller looks for instructions on what to do next.
- 2) A **Read/Write (R/W) pin** that selects reading mode or writing mode
- 3) An **Enable pin** that enables writing to the registers
- 4) **8 data pins (D0 -D7)**. The states of these pins (high or low) are the bits that you're writing to a register when you write, or the values you're reading when you read.

A) Also, the Liquid Crystal library shows how to use the `scrollDisplayLeft ()` and `scrollDisplayRight ()` methods to reverse the direction the text is flowing.

B) For custom character display,

Create a custom character for use on the LCD. Up to eight characters of 5x8 pixels are supported (numbered 0 to 7). The appearance of each custom character is specified by an array of eight bytes, one for each row. The five least significant bits of each byte determine the pixels in that row. To display a custom character on the screen, `write()` its number.

There's also a **display contrast pin (Vo)**, **power supply pins (+5V and Gnd)** and **LED Backlight (Bklt+ and Bklt-)** pins that you can use to power the LCD, control the display contrast, and turn on and off the LED backlight, respectively.



LCD functions: 1. LiquidCrystal. Begin(16x2)- Initializes LCD specifying size in number of columns and rows.

Electronic components and materials required:

- Arduino UNO Development Board,
- USB Cable,
- Breadboard,
- LED, Resistor (1 k ohm),
- 10K Potentiometer,
- Connecting Wires,
- 16x2 LCD Module.
- **I2C module**

ACTUAL CONNECTED CIRCUIT PHOTOS

Task 1

Circuit Connection photo before operation/ Working

Actual Photograph during Working of the circuit

EXPLANATION OF THE RESULT OF EACH TASK

Flowchart/ Algorithm: Write detailed steps of algorithm for each Tasks (A, B, C and D)

Step - 1

Open Arduino IDE. Here use "Liquid Crystal" library. This is an inbuilt library, so no need to install that library separately.

```
#include <LiquidCrystal.h>
```

```
#include <LiquidCrystal_I2C> (If I2C module is available
```

Step -2

Next initialize the library with the number of the interface pins. With the function "LiquidCrystal lcd ()". The function has six attributes. These are the interface pins in the order of "RS, E, D4, D5, D6, D7". Here we use pins 12, 11, 5, 4, 3, 2. corresponding to above.

```
LiquidCrystal lcd (12, 11, 5, 4, 3, 2);
```

Step - 3

Now can call this display by "lcd". Next program the setup part. We need to set the number of columns and number of rows. Here use the LCD with 16 column and 2 rows. Set the number of columns and rows by the function "lcd.begin(16, 2)". If you have a display with 16 columns and 4 rows this become "lcd.begin(16, 4)". And set the A0 pin as input.

```
lcd.begin(16,2);  
pinMode(A0,INPUT);
```

Step - 4

The setup part is over. Next loop part First to clear the display use the function "lcd. Clear ()". This function will clean the entire display.

Step - 5

We need a starting point to start printing. So set the cursor to that particular point by the function "lcd.setCursor()". This function has only two attributes. It is that starting points. (The number of column and number of row). Start from first column first row. The first column is represented as 0, second is 1, and so on and the first row is represented as 0, second is 1. So, we need to start from the position (0, 0). You can easily understand if you know about matrix. The piece of code become,

```
lcd.setCursor(0,0);
```

Step - 6

The instruction to print. to print "Hello World" by the instruction "lcd.print()". Alternatively, you can print anything. Please don't forgot the double quote marks.

```
lcd.print("Hello World");
```

Step -7

In the above printing statement, we use total of 14 characters. So, the current position of cursor is at (14, 0). Now to print on the next line, i.e., the position (0, 1). Set the cursor to that point by the function "lcd.setCursor()".

```
lcd.setCursor(0,1);
```

Step - 8

Now the cursor is at the position of second row and first column (0, 1). Then print another text "Value" by the function lcd.print().

```
lcd.print("Value : ");
```

Step - 9

Here we use 8 characters in second line. So, the current position of cursor is (8, 1). Next, we need to read the analog value from the pin A0 and print it to the display. It uses "analogRead()" function to read the analog value and use the "lcd.print()" function to print the value to the display. And there is no need of double quotes.

```
lcd.print(analogRead(A0));
```

Step - 10

Delay (): we have to add some delay. Otherwise, the text will blink continues. That because of the first instruction "LCD. Clear ()". Every time when see this command the Arduino will clear the display, This results the blinking of display. But use of delay will decrease this blinking.

```
Delay (500);
```

Step - 11



Connection

Vishwakarma Institute of Technology
Department of Engineering, Sciences and Humanities
FY: 2024-2025: Semester I & II
ET1012: Applied Electromechanics

- * LCD RS pin to digital pin 12
- * LCD Enable pin to digital pin 11
- * LCD D4 pin to digital pin 5
- * LCD D5 pin to digital pin 4
- * LCD D6 pin to digital pin 3
- * LCD D7 pin to digital pin 2
- * LCD R/W pin to ground
- * LCD VSS pin to ground
- * LCD VCC pin to 5V
- * 10K POT:
- * ends to +5V and ground
- * wiper to LCD VO pin (pin 3)

Conclusion:

LCDs are mostly used in portable electronic games, in flat panel televisions, in electronic billboards, in video- projection system and many more such devices



Experiment No. 10

Title of the assignment—_____

Name of the student—_____

Roll No.—_____ **Batch—**_____ **Division—**_____ **PR No.—**_____

Date of performance—_____

Experiment No. 10

Title: Thermistor analog temperature sensor interfacing with Arduino

Pre requisite:

Tasks:

A) Take temperature readings from the sensor and display the results on the Serial Monitor.

Theory: A Thermistor is a type of resistor whose resistance is dependent on temperature. There are two opposite types of thermistors:

PTC (Positive Temperature Coefficient): In this type of Thermistor the temperature increases with the decrease of the resistance. The resistance of the negative temperature coefficient Thermistor is very large due to which it detects the small variation in temperature (resistance increases as temperature rises)

NTC (Negative Temperature Coefficient): The resistance of the Thermistor increases with the increases in temperature. (Resistance decreases as temperature rises)

Note that the sensor operates on a voltage range of 4 to 30 V and that the output voltage is independent of the supply voltage.

Operating components and materials required with specifications:

- Arduino Development Board,
- USB Cable,
- Breadboard,
- Thermistor analog temperature sensor
- Connecting Wires,

Measure Temperature using a Thermistor and Arduino Uno:

Here, an NTC type Thermistor of 10kΩ is used. NTC of 10kΩ means that this Thermistor has a resistance of 10kΩ at 25°C. Voltage across the 10kΩ resistor is given to the ADC of UNO board. The Thermistor resistance is found out using simple voltage divider network formula

$$R_{th} + 10\text{ k} = \frac{5 * 10\text{ k}}{V_{out}}$$

R_{th} is the resistance of thermistor.

$$V_{out} = 5 * \frac{10\text{ k}}{10\text{ k} + R_{th}}$$

V_{out} is the voltage measured by the ADC

$$V_{out} = \frac{5 * ADC\ val}{1023}$$

V_{out} =



The temperature can be found out from thermistor resistance using the Steinhart-Hart equation.

$$\text{Temperature in Kelvin} = 1 / (A + B[\ln(R)] + C[\ln(R)]^3)$$

where $A = 0.001129148$, $B = 0.000234125$ and $C = 8.76741 \times 10^{-8}$, and R is the thermistor resistance.

ACTUAL CONNECTED CIRCUIT PHOTOS

Task 1

Circuit Connection photo before operation/ Working

Actual Photograph during Working of the circuit

Photograph of Serial Monitor depicting the result

EXPLANATION OF THE RESULT OF EACH TASK

Conclusion:



Experiment No. 11

Title of the assignment—_____

Name of the student—_____

Roll No.—_____ **Batch—**_____ **Division—**_____ **PR No.—**_____

Date of performance—_____

Experiment No. 11

Title: IR sensor interfacing with Arduino Uno

Prerequisites: Various Sensors, and its usage in applications, Sensitivity Control of Sensor.

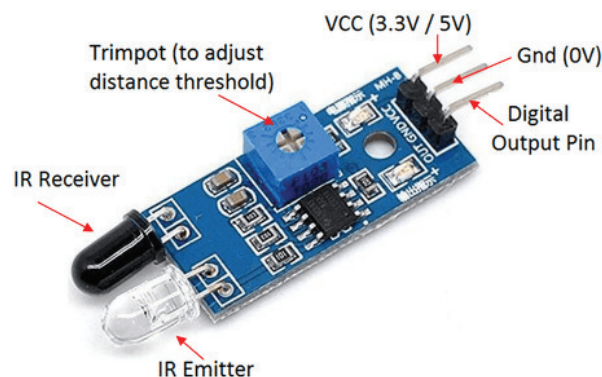
Problem statement

- A) Write a program to interface **IR sensor** so as to display the presence of obstacle on serial monitor.

Theory: An IR sensor, also known as an Infrared sensor, is an electronic device that detects and measures infrared radiation. Infrared radiation is a type of electromagnetic radiation with a wavelength longer than visible light. IR sensors work by emitting an infrared beam and then measuring the amount of reflected light. They are widely used in various applications such as remote-control devices, security systems, obstacle detection, and industrial automation. The IR sensor consists of two main parts: an infrared emitter and an infrared detector. The emitter emits an infrared beam, which bounces back to the detector if it encounters an object. The detector measures the reflected light and converts it into an electrical signal, which is then processed by the sensor's circuit. Based on the electrical signal, the IR sensor can determine the distance, speed, or presence of an object. IR sensors are also available in both analog and digital forms, with different specifications for sensitivity, accuracy, and response time.

IR Sensor Module

IR sensor module hardware mainly includes IR LED/IR Tx, Photodiode/IR Rx, LM393 comparator IC, signal LED, power LED, Trim pot, and pins like VCC, GND & Output



Operating components and materials required with specifications:

- Arduino Development Board,
- USB Cable,

- Breadboard,
- Connecting Wires,
- IRModule

Interfacing Circuit diagram: (Draw Connections between Arduino Uno and the IR Sensor as well as other components)



Code : (Put screenshot here)

Result:(Put screenshot of serial monitor for each problem statement given i.e., for each of the Aim A) and B).

Applications of IR Sensor:

Proximity Sensing: IR sensors are commonly used for proximity sensing applications, such as detecting the presence of an object or person in close proximity to the sensor for e.g., in Automatic Hand Sanitizer.

Motion Detection: IR sensors can be used as motion detectors to sense movement in an area. They are often used in security systems, automatic lighting controls, and burglar alarms. PIR motion sensor is used in these types of Application.

Remote Control Systems: IR sensors are extensively used in remote control devices, such as TVs, air conditioners, and home entertainment systems. They receive infrared signals from remote controllers and convert them into electrical signals for device control.

Ambient Light Sensing: IR sensors can measure ambient light levels and adjust the brightness of displays or lighting systems accordingly. They are commonly found in smartphones, tablets, and automatic lighting systems.

Object Counting: IR sensors can be used to count the number of objects passing through a specific area, such as people entering or exiting a building or items on a conveyor belt.

Conclusion:



Experiment No. 12

Title of the assignment—_____

Name of the student—_____

RollNo.—_____ **Batch—**_____ **Division—**_____ **PRNo.—**_____

Date of performance —_____

Experiment No. 12 Title: Seven Segment Display

Interfacing with Arduino (4 digit 7 segment module)

Pre-requisite: LED Display Construction and working, seven segment display code concept,

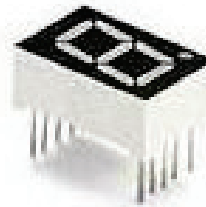
Counter Operation

Tasks

- A) Write a program to turn ON and turn OFF the individual LED.
- B) Write a program to implement simple decimal counter (Counting 0 to 9) using a 7 Segment Display.

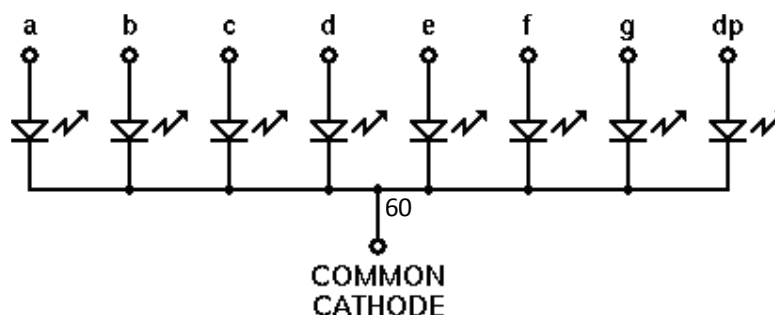
Theory:

4 Digit seven segment LED module . We need to download and install library for driver TM1637

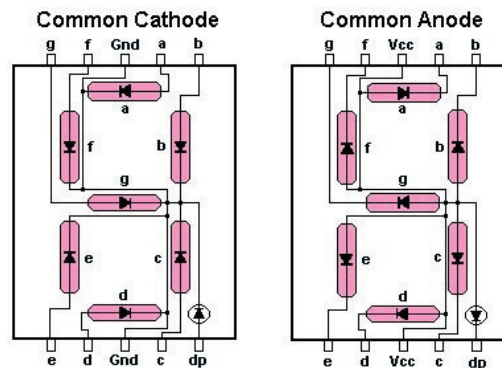
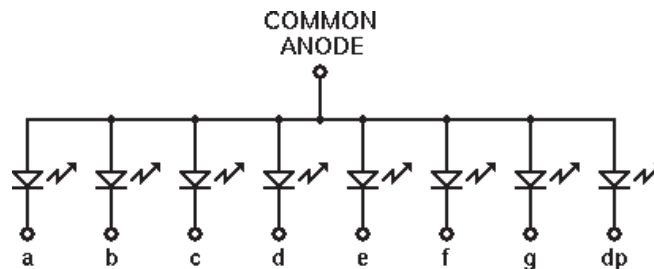


When Arduino application only needs to display numbers, consider using a seven-segment display. The seven-segment display has seven LEDs arranged in the shape of number eight. They are easy to use and cost effective.

Seven segment displays are of two types: common anode and common cathode. The Internal structure of both types is nearly the same. The difference is the polarity of the LEDs and common terminal. In a common cathode seven-segment display (most popularly used), all seven LEDs plus a dot LED have the cathodes connected to pins 3 and pin 8. To use this display, we need to connect GROUND to pin 3 and pin 8 and, and connect +5V to the other pins to make the individual segments light up. The following diagram shows the internal structure of common-cathode seven-segment display:



The common anode display is the exact opposite. In a common-anode display, the positive terminal of all the eight LEDs is connected together and then connected to pin 3 and pin 8. To turn on an individual segment, you ground one of the pins. The following diagram shows the internal structure of the common-anode seven-segment display.



Operating components and materials required with specifications:

- Arduino UNO Development Board,
- USB Cable, Breadboard,
- Connecting Wires,
- Push Button (Micro Switch)
- Segment Display (Common Cathode Type) or 4 digit module



Interfacing Circuit diagram: (Draw Connections between Arduino Uno and the display as well as other components)

Code: (Put Screenshot of code) :



Result : (Put screenshot of serial monitor for each problem statement given i.e., for each of the Aim A) and B) and C)

Applications:

- 1) Digital Displays of Testing Machines
- 2) Digital Clock

Conclusion:



Experiment No. 13

Title of the assignment—_____

Name of the student—_____

Roll No.—_____ **Batch—**_____ **Division—**_____ **PRNo.—**_____

Date of performance—_____

Experiment No. 13

Title: Servo Motor Interfacing with Arduino

Pre-requisite: Difference between servo and DC motor, PWM, Current requirements of a Servo

Tasks:

- A) Write a program to control the Servo Motor Rotation by one Degree using Arduino Uno
- B) Write a program to control Servo Motor Rotation Control using potentiometer.
- C) **Write a program to rotate servo through given degrees. Accept Command from cell phone via blue tooth.**

Theory:

A servo motor is an electric device used for precise control of angular rotation. It is used in applications that demand precise control over motion, like in case of control of a robotic arm, The rotation angle of the servo motor is controlled by applying a PWM signal to it. By varying the width of the PWM signal, we can change the rotation angle and direction of the motor.

Servo motors have three wires: power, ground, and signal. The power wire is typically red, and should be connected to the 5V pin on the Arduino board. The ground wire is typically black or brown and should be connected to a ground pin on the board. The signal pin is typically yellow or orange and should be connected to PWM pin on the board. In these examples, it is pin number 9.

The Servo.h is a great library for controlling servo motors.

Knob Circuit

For the Knob example, wire the potentiometer so that its two outer pins are connected to power (+5V) and ground, and its middle pin is connected to A0 on the board. Then, connect the servo motor to +5V, GND and pin 9.

Functions Used

1. Servo my servo
 - This creates an object named my servo of the class Servo.
2. *myservo.attach(pin)*
 - This function attaches the servo variable to a pin.
 - Pin is the pin number to which the servo is attached.
3. *myservo.write(angle)*
 - This function writes a value to the servo, thus controlling the position of the shaft.
 - Angle can take values between 0 to 180.
4. *map (value, fromLow, fromHigh, toLow, toHigh)*

- This function is used to map a number from one range to another range.
- This means that “value” having a value between “fromLow” to “fromHigh” gets converted to equivalent values in the range of “toLow” to “toHigh”. “fromLow” gets mapped to “toLow” and so on.



Operating components and materials required with specifications:

- Arduino UNO Development Board,
- USB Cable, Breadboard,
- Connecting Wires,
- 10 K Potentiometer
- Servo

Interfacing Circuit diagram: (Draw Connections between Arduino Uno and the Sensors as well as other components)

Algorithm: Write (Type) detailed steps of algorithm for each Tasks given i.e., for each of the Aim A), & B).

A) Program to control the Servo Motor Rotation by one Degree

STEP 1-Start

STEP 2- include servo.h library

STEP 3- define angle

STEP 4- create a new object myservo from the library

STEP 5- in void setup, declare myservo.attach(9)

STEP 6- in void loop, use for loops to rotate the motor

STEP 7-Stop



B) Program to control servo motor with potentiometer.

STEP 1-Start

STEP 2- include servo.h library

STEP 3- create a new object myservo from the class Servo

STEP 4- define a variable val and port pin and initialize it to the analogue pin of the Arduino A0

STEP 5- in void setup, declare myservo.attach(9)

STEP 6- in void loop, initialize the variable val to the value from the analogue input of the potentiometer

STEP 7- map the value of angle and rotate the motor

STEP 8- Stop

Result: Put screenshot of serial monitor for each Tasks given i.e., for each of the Aim A), & B).

Applications:

Servo motors are used in DVD and Blue ray disc players. Automobiles also use servo motors. In modern cars it is used to control its speed. Printers also uses the servo.

Conclusion:



Experiment No. 14

Title of the assignment—_____

Name of the student—_____

Roll No.—_____ **Batch—**_____ **Division—**_____ **PRNo.—**_____

Date of performance—_____

Experiment No. 14

Title: DC Motor Interfacing with Arduino

Pre-requisite: PWM, Current requirements of DC Motor, Calculation of required voltage and Current of a DC motor.

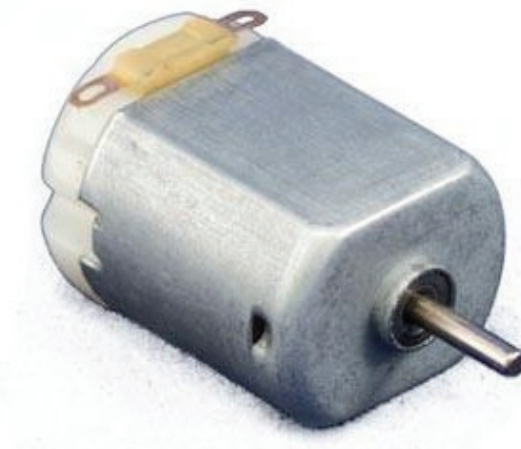
Tasks

- A) Write a program to control DC motor clockwise & anti clockwise rotation control using a push button.
- B) Write a program to control Speed of DC Motor Rotation using a potentiometer or H bridge.
- C) Write a program to drive DC motor using Relay.

Components: Arduino UNO Development Board, USB Cable, Breadboard, LED, Resistor (1 k ohm), Connecting Wires, Potentiometer 10K, DC Motor relay.

Theory:

DC motor converts electrical energy in the form of Direct Current into mechanical energy in the form of rotational motion of the motor shaft. The DC motor speed can be controlled by applying varying DC voltage; whereas the direction of rotation of the motor can be changed by reversing the direction of current through it. For applying varying voltage, we can make use of PWM technique. For reversing the current, we can make use of H-Bridge circuit or motor driver ICs that employ the H-Bridge technique. PWM wave generated on the Arduino UNO can be used to provide a variable voltage to the motor through L293D. In Arduino, analog. Write function is used to generate PWM wave.



1. analogWrite (pin, value)

- This function is can be used for generating PWM on PWM digital pins (3, 5,6,9,10,11 for Arduino UNO).
- Value can be any number between 0 to 255. 0 being 0% duty cycle and 255 being 100% duty cycle.

Operating components and materials required with specifications:

- Arduino UNO Development Board,
- USB Cable, Breadboard,
- Connecting Wires,
- 10 K Potentiometer
- DC Motor 12V 200 rpm



Interfacing Circuit diagram: (Draw Connections between Arduino Uno and the motor as well as other components)

Result: Put screenshot of serial monitor if applicable

Conclusion:
